T-Test (Student's t-Distribution)

The **T-test** is a type of inferential statistic used to determine if there is a significant difference between the means of two groups, or between a sample mean and a known population mean, especially when the sample size is small.

T-Test Formula (One-Sample)

The formula for the one-sample T-test is structurally similar to the Z-test, but it uses the **sample standard deviation (S)** to estimate the population standard deviation (σ).

$$t_{\text{Calculated}} = \frac{\bar{x} - \mu_0}{\frac{S}{\sqrt{n}}}$$

Where:

- $\bar{x}$: **Sample Mean**
- μ_0: **Hypothesized Population Mean** (from H_0)
- S: **Sample Standard Deviation** (used instead of the unknown σ)
- n: **Sample Size**

When to Use the T-Test

The T-test is primarily used for hypothesis testing when the conditions for the Z-test (large sample or known population standard deviation) are **not met**.

1. **Small Sample Size:** The sample size (n) is **less than 30** ($n < 30$).
2. **Unknown Population Standard Deviation:** The true population standard deviation (σ) is **unknown**.

> !! **Rationale:** When n is small and σ is unknown, substituting the sample standard deviation (S) for σ introduces more uncertainty and variability into the test statistic. The **t-distribution** is wider (flatter tails) than the Normal (Z) distribution, which accounts for this increased uncertainty, making it a more conservative and appropriate test.

The T-Distribution and Degrees of Freedom (df)

Explanation of Degrees of Freedom (df)

The **t-distribution** is not a single curve but a family of curves. The exact shape of the curve depends on a value called the **degrees of freedom (df)**.

- **Degrees of Freedom (df):** The number of independent values that can vary in a data set.
- **Why $n-1$ for one-sample:** In the one-sample T-test, we use the sample mean ($\bar{x}$) to calculate the sample standard deviation (S). Once $\bar{x}$ is fixed, only $n-1$ scores in the sample can vary freely. The last score is fixed because it must satisfy the calculated mean.
  - df for one-sample t-test is: $df = \mathbf{n-1}$.

Information Conveyed by the T-Statistic

The t-statistic is a measure of the difference between the sample mean and the hypothesized population mean, scaled by the estimated standard error.

- $$t = \frac{\text{Observed Difference}}{\text{Estimated Standard Error}}$$

1. **Numerator ($\bar{x} - \mu_0$):** Measures the magnitude of the difference between your sample data ($\bar{x}$) and the null hypothesis claim (μ_0).
2. **Denominator ($\frac{s}{\sqrt{n}}$):** Measures the estimated standard error, which is the amount of difference expected by **chance** (random sampling error).

If the **$|t|$-statistic** is large, it means the observed difference is much greater than what is expected by chance, leading to the **rejection of H_0**.

Calculating Degrees of Freedom and Critical Values

Test Type	Degrees of Freedom (df) Formula	Example ($n_1 = 8, n_2 = 12$)	Locating Critical Values
One-Sample t-Test (Testing $\bar{x}$ vs. μ_0)	$df = \mathbf{n - 1}$	If $n = 10$, $df = 10 - 1 = 9$.	Locate the df row (e.g., $df = 9$) and the α column (e.g., $\alpha = 0.05$ two-tailed) to find the critical t-value in the **t-table**.
Two-Independent-Sample t-Test (Testing $\bar{x}_1$ vs. $\bar{x}_2$)	$df = \mathbf{n_1 + n_2 - 2}$	$df = 8 + 12 - 2 = 18$.	Locate the $df = 18$ row and the appropriate α column.

Assumptions for the T-Tests

All T-tests rely on certain assumptions about the population from which the samples are drawn. If these assumptions are seriously violated, the results of the test may not be reliable.

Assumptions for the One-Sample T-Test

1. **Random Sampling:** The sample must be drawn using a **random sampling** method. This ensures the sample is representative of the population.
2. **Independence of Observations:** Each observation (data point) within the sample must be independent of all other observations.
3. **Normality:** The population from which the sample is drawn must be **normally distributed**.
   - *Note:* This assumption becomes less critical as the sample size (n) increases (due to the Central Limit Theorem), but it is important for small samples.

Assumptions for the Two-Independent-Sample T-Test

This test is used to compare the means of two distinct, unrelated groups (e.g., comparing test scores of males vs. females).

1. **Independence of Observations (within and between groups):** Observations within each group must be independent, and the two groups must be independent of each other (i.e., the individuals in Group 1 have no connection to the individuals in Group 2).
2. **Normality:** Both populations from which the two samples are drawn must be **normally distributed**.
3. **Homogeneity of Variances:** The variance (or standard deviation) of the population for Group 1 must be **equal** to the variance of the population for Group 2 ($\sigma_1^2 = \sigma_2^2$).
   - If this assumption is violated, a modified, more complex formula (unequal variances t-test) must be used.

Hypothesis Testing (t-test/Small Sample Tests) Examples

We will use the **t-test** for the first set of examples because the sample sizes (n) are generally small ($n < 30$) and the population standard deviation (σ) is unknown, requiring us to use the sample standard deviation (S).

Example 1: One-Sample t-Test (Calculating $\bar{x}$ and S)

Question: The **9 items** of a sample have the values **45, 47, 50, 52, 48, 47, 49, 53, 51**. Does the mean of these values differ significantly from the assumed mean **47.5**. (Test at **5% level of significance**)

- $n = 9$ (Small Sample)
- Assumed Population Mean: $\mu_0 = 47.5$
- $\alpha = 0.05$ (Two-Tailed Test, as we check if it "differs")

1. Calculate Sample Mean ($\bar{x}$) and Sample Standard Deviation (S)

- $\sum x = 45 + 47 + 50 + 52 + 48 + 47 + 49 + 53 + 51 = 442$
- $\bar{x} = \frac{\sum x}{n} = \frac{442}{9} \approx 49.111$
- $\sum (x - \bar{x})^2$:
  - $(45 - 49.11)^2 + (47 - 49.11)^2 + \cdots + (51 - 49.11)^2 \approx 42.889$
- **Sample Variance (S^2):** $S^2 = \frac{\sum (x-\bar{x})^2}{n-1} = \frac{42.889}{8} \approx 5.361$
- **Sample Standard Deviation (S):** $S = \sqrt{5.361} \approx 2.315$

2. Hypotheses and Critical Value

- H_0: $\mu = 47.5$
- H_1: $\mu \neq 47.5$
- **Degrees of Freedom:** $df = n - 1 = 9 - 1 = 8$

- **Critical Value (t_{Critical}):** For $df = 8$ and $\alpha = 0.05$ (Two-Tailed), $t_{\text{Critical}} = \pm 2.306$.

3. Calculate the t-Statistic

$$t_{\text{Calculated}} = \frac{\bar{x} - \mu_0}{\frac{S}{\sqrt{n}}} = \frac{49.111 - 47.5}{\frac{2.315}{\sqrt{9}}} = \frac{1.611}{\frac{2.315}{3}} = \frac{1.611}{0.7717} \approx 2.0876$$

4. Conclusion

- $|t_{\text{Calculated}}| = 2.0876$
- $|t_{\text{Critical}}| = 2.306$
- Since $2.0876 < 2.306$, the calculated t falls within the acceptance region.
- **Result: Accept H_0** (Fail to Reject H_0).
- **Interpretation:** The mean of the sample values ($\bar{x} \approx 49.11$) **does not differ significantly** from the assumed mean of 47.5 at the 5% level.

Example 2: One-Sample t-Test (Given $\bar{x}$ and S)

Question: A sample of **20 items** has mean **42 units** and S.D. **5 units**. Test the hypothesis that it is a random sample from a normal population with mean **45 units**. (Test at **5% level of significance**)

- $n = 20$ (Small Sample)
- Sample Mean: $\bar{x} = 42$
- Sample Standard Deviation: $S = 5$
- Hypothesized Population Mean: $\mu_0 = 45$
- $\alpha = 0.05$ (Two-Tailed Test, testing if it is "a random sample from")

1. Hypotheses and Critical Value

- H_0: $\mu = 45$
- H_1: $\mu \neq 45$
- **Degrees of Freedom:** $df = n - 1 = 20 - 1 = 19$
- **Critical Value (t_{Critical}):** For $df = 19$ and $\alpha = 0.05$ (Two-Tailed), $t_{\text{Critical}} = \pm 2.093$.

2. Calculate the t-Statistic

$$t_{\text{Calculated}} = \frac{\bar{x} - \mu_0}{\frac{S}{\sqrt{n}}} = \frac{42 - 45}{\frac{5}{\sqrt{20}}} = \frac{-3}{\frac{5}{4.472}} = \frac{-3}{1.118} \approx -2.683$$

3. Conclusion

- $|t_{\text{Calculated}}| = 2.683$
- $|t_{\text{Critical}}| = 2.093$
- Since $2.683 > 2.093$, the calculated t falls into the rejection region.
- **Result: Reject H_0.**
- **Interpretation:** The sample mean of 42 is **significantly different** from the hypothesized population mean of 45 at the 5% level. It is **not** likely a random sample from that population.

Example 3: One-Sample t-Test (Calculating $\bar{x}$ and S, One-Tailed)

Question: The height of **8 males** are **175, 168, 165, 170, 167, 160, 173 and 168 cm**. Can we conclude that the avg. height is **greater than 165 cm**. (Test at **5% level of significance**)

- $n = 8$ (Small Sample)
- Hypothesized Population Mean: $\mu_0 = 165$
- $\alpha = 0.05$ (Right-Tailed Test, checking if "greater than")

1. Calculate Sample Mean ($\bar{x}$) and Sample Standard Deviation (S)

- $\sum x = 175 + 168 + 165 + 170 + 167 + 160 + 173 + 168 = 1346$
- $\bar{x} = \frac{1346}{8} = 168.25$ cm
- $\sum(x - \bar{x})^2 \approx 135.5$
- **Sample Standard Deviation (S):** $S = \sqrt{\frac{135.5}{8-1}} = \sqrt{19.357} \approx 4.3996$

2. Hypotheses and Critical Value

- **H_0:** $\mu = 165$
- **H_1:** $\mu > 165$ (Right-Tailed Test)
- **Degrees of Freedom:** $df = 8 - 1 = 7$
- **Critical Value (t_{Critical}):** For $df = 7$ and $\alpha = 0.05$ (One-Tailed), $t_{\text{Critical}} = +1.895$.

3. Calculate the t-Statistic

$$t_{\text{Calculated}} = \frac{\bar{x} - \mu_0}{\frac{S}{\sqrt{n}}} = \frac{168.25 - 165}{\frac{4.3996}{\sqrt{8}}} = \frac{3.25}{\frac{4.3996}{2.828}} = \frac{3.25}{1.555} \approx 2.090$$

4. Conclusion

- $t_{\text{Calculated}} = 2.090$
- $t_{\text{Critical}} = 1.895$
- Since $2.090 > 1.895$, the calculated t falls into the rejection region.
- **Result: Reject H_0.**
- **Interpretation:** We **can conclude** that the average height of the males is **greater than 165 cm** at the 5% level.

Example 4: Two-Sample Z-Test (Large Sample Formula Used Due to Known σ)

Question: Samples of size **10 and 14** were taken from two normal populations with S.D. **3.5 and 5.2**. The sample means were found to be **20.3 and 18.6**. Test that whether the means of two populations are the same at **5% level**.

- $n_1 = 10, n_2 = 14$
- $\bar{x}_1 = 20.3, \bar{x}_2 = 18.6$
- **Crucial Point:** The population standard deviations ($\sigma_1 = 3.5$ and $\sigma_2 = 5.2$) are **known**. Even though $n < 30$, when population σ is known, the Z-test is the correct procedure.

1. Hypotheses and Critical Value

- H_0: $\mu_1 = \mu_2$
- H_1: $\mu_1 \neq \mu_2$ (Two-Tailed)
- **Critical Value (Z_{Critical}):** For $\alpha = 0.05$ (Two-Tailed Z-test), $Z_{\text{Critical}} = \pm 1.96$.

2. Calculate the Z-Statistic

$$Z_{\text{Calculated}} = \frac{(\bar{x}_1 - \bar{x}_2) - 0}{\sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}} = \frac{20.3 - 18.6}{\sqrt{\frac{3.5^2}{10} + \frac{5.2^2}{14}}}$$

$$Z = \frac{1.7}{\sqrt{\frac{12.25}{10} + \frac{27.04}{14}}} = \frac{1.7}{\sqrt{1.225 + 1.931}} = \frac{1.7}{\sqrt{3.156}} = \frac{1.7}{1.7765} \approx 0.9569$$

3. Conclusion

- $|Z_{\text{Calculated}}| = 0.9569$
- $|Z_{\text{Critical}}| = 1.96$
- Since $0.9569 < 1.96$, the calculated Z falls within the acceptance region.
- **Result: Accept H_0** (Fail to Reject H_0).
- **Interpretation:** There is **no significant difference** between the means of the two populations.

Example 5: Two-Independent-Sample t-Test (Calculating $\bar{x}$ and Pooled S)

Question: The height of **6 sailors** are **63, 65, 68, 69, 71 and 72**. Those of **9 soldiers** are **61, 62, 65, 66, 69, 70, 71, 72 and 73**. Test whether the sailors are on the avg. taller than soldiers. (Test at **5% level of significance**)

This requires the two-sample t-test using a **pooled variance** because σ is unknown and sample sizes are small.

1. Calculate Sample Statistics

Group	n	$\sum x$	$\bar{x}$
Sailors (1)	$n_1 = 6$	408	$\bar{x}_1 = 68$
Soldiers (2)	$n_2 = 9$	650	$\bar{x}_2 = 72.22$
Recalculation for Soldiers: $\bar{x}_2 = \frac{61+62+65+66+69+70+71+72+73}{9} = \frac{609}{9} \approx 67.67$	$\sum(x_2 - \bar{x}_2)^2 \approx 135.56$		

*Correction based on sample data provided: $\sum x_2 = 609$, $\bar{x}_2 = 67.67$.
$\sum(x_2 - \bar{x}_2)^2 \approx 135.56$.

2. Hypotheses and Critical Value

- H_0: $\mu_1 = \mu_2$ (Sailors are not taller or are the same height.)
- H_1: $\mu_1 > \mu_2$ (Sailors are taller - Right-Tailed Test)
- **Degrees of Freedom:** $df = n_1 + n_2 - 2 = 6 + 9 - 2 = 13$
- **Critical Value (t_{Critical}):** For $df = 13$ and $\alpha = 0.05$ (One-Tailed), $t_{\text{Critical}} = +1.771$.

3. Calculate the t-Statistic

A. Pooled Standard Deviation (S_p):

$$S_p^2 = \frac{(n_1-1)S_1^2 + (n_2-1)S_2^2}{n_1+n_2-2} = \frac{\sum(x_1-\bar{x}_1)^2 + \sum(x_2-\bar{x}_2)^2}{n_1+n_2-2}$$

$$S_p^2 = \frac{50+135.56}{13} = \frac{185.56}{13} \approx 14.274$$

$$S_p = \sqrt{14.274} \approx 3.778$$

B. $t_{\text{Calculated}}$:

$$t_{\text{Calculated}} = \frac{\bar{x}_1 - \bar{x}_2}{S_p\sqrt{\frac{1}{n_1}+\frac{1}{n_2}}} = \frac{68-67.67}{3.778\sqrt{\frac{1}{6}+\frac{1}{9}}}$$

$$t = \frac{0.33}{3.778\sqrt{0.1667+0.1111}} = \frac{0.33}{3.778\sqrt{0.2778}} = \frac{0.33}{3.778\cdot 0.527} = \frac{0.33}{1.991} \approx 0.166$$

4. Conclusion

- $t_{\text{Calculated}} = 0.166$
- $t_{\text{Critical}} = 1.771$
- Since $0.166 < 1.771$, the calculated t falls within the acceptance region.
- **Result: Accept H_0** (Fail to Reject H_0).
- **Interpretation:** There is **no significant evidence** to conclude that the sailors are, on average, taller than the soldiers.

Quiz and Assignment Questions Solutions

Quiz Question 1: Two-Sample Z-Test (Known σ)

- $Z_{\text{Calculated}} \approx 0.9569$ (Calculated in Example 4).
- $Z_{\text{Tabulated}} = 2.0739$ (This is a t-table value, but since σ is known, we should use the Z-critical value of 1.96 for $\alpha = 0.05$). Assuming the question intended to provide the

correct t-critical value for $df = 22$ (for t-test) or the intended Z-critical value 1.96:

- If using $Z_{\text{Crit}} = 1.96$: $0.9569 < 1.96$. **Accept H_0**.
- If using $t_{\text{Crit}} = 2.0739$ (ignoring the known σ): $0.9569 < 2.0739$. **Accept H_0**.

- **Result:** The means of the two populations are considered **the same**.

Quiz Question 2: Two-Sample Z-Test (Large Samples)

	Mean ($\bar{x}$)	S.D. (S)	Size (n)
Girl (1)	75	8	60
Boys (2)	73	10	100

- Both n_1 and n_2 are large, so use the **Z-test**.
- $H_0 : \mu_1 = \mu_2$, $H_1 : \mu_1 \neq \mu_2$. $Z_{\text{Critical}} = \pm 1.96$ $(\alpha = 0.05)$.
- $Z_{\text{Calculated}}$:

$$Z = \frac{75 - 73}{\sqrt{\frac{8^2}{60} + \frac{10^2}{100}}} = \frac{2}{\sqrt{1.0667 + 1.0}} = \frac{2}{\sqrt{2.0667}} = \frac{2}{1.4376} \approx 1.391$$

- **Conclusion:** $|1.391| < 1.96$. **Accept H_0**.
- **Answer:** The difference between the mean scores is **not statistically significant**.

Quiz Question 3: One-Sample t-Test (One-Tailed)

- $n = 25$ (Small Sample), $\bar{x} = 80$, $\mu_0 = 50$, $S = 15$.
- $H_0 : \mu = 50$, $H_1 : \mu > 50$ (Right-Tailed Test).
- $df = 25 - 1 = 24$. t_{Critical} for $df = 24, \alpha = 0.05$ (One-Tailed) $= +1.711$.
- $t_{\text{Calculated}}$:

$$t = \frac{80 - 50}{\frac{15}{\sqrt{25}}} = \frac{30}{\frac{15}{5}} = \frac{30}{3} = \mathbf{10.0}$$

- **Conclusion:** $10.0 > 1.711$. **Reject H_0**.
- **Answer:** Yes, the training has **significantly improved** the sales.

Quiz Question 4: Two-Sample Z-Test (Large Samples)

	Mean ($\bar{x}$)	S.D. (S)	Size (n)
Girl (1)	68	5	60
Boys (2)	85	10	100

- Both n_1 and n_2 are large, so use the **Z-test**.
- $H_0 : \mu_1 = \mu_2$, $H_1 : \mu_1 \neq \mu_2$. $Z_{\text{Critical}} = \pm 1.96$ $(\alpha = 0.05)$.
- **$Z_{\text{Calculated}}$:**

$$Z = \frac{68 - 85}{\sqrt{\frac{5^2}{60} + \frac{10^2}{100}}} = \frac{-17}{\sqrt{0.4167 + 1.0}} = \frac{-17}{\sqrt{1.4167}} = \frac{-17}{1.190} \approx -14.286$$

- **Conclusion:** $|-14.286| > 1.96$. **Reject H_0**.
- **Answer:** The difference between the mean scores is **highly significant**.

Multiple Choice Questions (MCQs) Solutions

- **A sample of 20 items...**: This is **Example 2**.
  - **$t_{\text{Calculated}} \approx -2.683$**.
  - $|t_{\text{Calculated}}| = 2.683$.
  - Given $t_{\text{tab}} = 2.09$ (the critical value).
  - Since $2.683 > 2.09$, the result is that the null hypothesis is **Rejected**.
- **While testing the significance of difference of two sample means in case of small sample, how is degree of freedom calculated?**
  - The degrees of freedom for a two-independent-sample t-test is calculated as:

$$df = \mathbf{n_1 + n_2 - 2}$$